

CSE 400: Fundamentals of Probability in Computing

Lecture 26: Stochastic Processes (SSS, WSS, Ergodic)

1. Basics of Stochastic Processes [Revision]

1.1 Collection of Waveforms

Stochastic Process Definition

A stochastic process $X(\zeta, t)$ is an entire collection of waveforms or functions over time dependent on the outcome of a random experiment.

- The random output heavily depends on the experiment.

Example

Tossing a coin gives heads or tails resulting in specific deterministic waveforms.

1.2 Realizations and Random Variables

Fixed in Time Representation

Step 1:

A specific measured waveform from the experiment is a **Realization** of the process.

Step 2:

If we freeze the process at a specific time $t = t_1$, the process output evaluates to a single Random Variable:

$$X(\zeta, t_1) = X(t_1)$$

Step 3:

By evaluating at multiple time instances (t_1, t_2) , we acquire multiple dependent random variables:

$$X(t_1), X(t_2)$$

Step 4:

The dependency between these generated random variables dictates the fundamental nature of the stochastic process.

1.3 Characterization of Stochastic Process**Probability Density Functions (PDF)****Step 1: 1st Order Characterization**

$$t_1 \rightarrow f_{X_1}(x, t_1) \rightarrow F_{X_1}(x, t_1)$$

Step 2: 2nd Order Characterization

$$(t_1, t_2) \rightarrow f_{X_1 X_2}(x_1, x_2, t_1, t_2)$$

Step 3: nth Order Characterization

$$(t_1, \dots, t_n) \rightarrow f_{X_1, \dots, X_n}(x_1, \dots, x_n, t_1, \dots, t_n)$$

2. Statistical Moments**2.1 Mean Vector (First Order Moment)****Step 1: Definition**

$$\mu_X(t) = E[X(t)]$$

Step 2: Computation using PDF

$$\mu_X(t) = \int_{-\infty}^{\infty} x \cdot f_X(x, t) dx$$

Step 3: Property

$\mu_X(t)$ changes as a function of time

2.2 Autocorrelation Function**Step 1: Definition**

$$R_{XX}(t_1, t_2) = E[X(t_1)X^*(t_2)]$$

Step 2: Using Joint Density (2nd Order Characterization)

$$R_{XX}(t_1, t_2) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x_1 x_2 \cdot f_{X_1 X_2}(x_1, x_2, t_1, t_2) dx_1 dx_2$$

2.3 Properties of Autocorrelation

Conjugate Symmetry

Step 1: Property

$$R_{XX}(t_1, t_2) = R_{XX}^*(t_2, t_1)$$

Step 2: Proof Sketch

$$R_{XX}^*(t_2, t_1) = (E[X(t_2)X^*(t_1)])^*$$

$$(E[X(t_2)X^*(t_1)])^* = E[X^*(t_2)X(t_1)]$$

$$E[X^*(t_2)X(t_1)] = E[X(t_1)X^*(t_2)] = R_{XX}(t_1, t_2)$$

Positive Definiteness

Autocorrelation function forms a positive definite matrix.

2.4 Covariance Function

Definition

$$C_{XX}(t_1, t_2) = E[(X(t_1) - \mu_X(t_1))(X^*(t_2) - \mu_X^*(t_2))]$$

Normalized Form

$$\rho_{XX} = \frac{C_{XX}(t_1, t_2)}{\sigma_X \sigma_X}$$

2.5 Positive Definite Function

Quadratic Form Definition

$$\sum_{i=1}^n \sum_{j=1}^n a_i a_j^* R_{XX}(t_i, t_j) \geq 0$$

Matrix Expansion

$$[a_1^*; a_2^*; \dots; a_n^*] \begin{bmatrix} R_{11} & R_{12} & \dots \\ R_{21} & R_{22} & \dots \\ \vdots & \vdots & \ddots \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ \vdots \end{bmatrix} > 0$$

Equivalent Form

$$a^H R a > 0$$

2.6 Positive Definite Matrices

Conditions

1. A symmetric matrix is positive definite if all eigenvalues are positive.
2. A matrix is positive definite if all leading principal minors are positive:

$$|A_1| > 0, |A_2| > 0, \dots, |A_n| > 0$$

Example (Step-by-Step)

$$A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$$

Step 1:

$$|A_1| = 2 > 0$$

Step 2:

$$|A_2| = \begin{vmatrix} 2 & -1 \\ -1 & 2 \end{vmatrix} = 4 - 1 = 3 > 0$$

Step 3:

$$\begin{aligned} |A_3| &= 2(4 - 1) + 1(-2 - 0) \\ &= 6 - 2 = 4 > 0 \end{aligned}$$

Matrix is positive definite.

3. Stationary Stochastic Process

3.1 Strict Sense Stationary (SSS)

Step 1: First Order

$$f_X(x, t) = f_X(x, t + \tau) = g(x)$$

$$\mu_X(t) = \int xg(x) dx = \mu_X = \text{constant}$$

Step 2: Second Order

$$f_X(x_1, x_2; t_1, t_2) = f_X(x_1, x_2; t_1 + \tau, t_2 + \tau)$$

Choose:

$$\tau = -t_2$$

$$f_X(x_1, x_2; t_1, t_2) = f_X(x_1, x_2; t_1 - t_2, 0)$$

Step 3: Autocorrelation Derivation

$$\begin{aligned} R_{XX}(t_1, t_2) &= \int \int x_1 x_2 f_X(x_1, x_2; t_1, t_2) dx_1 dx_2 \\ &= \int \int x_1 x_2 f_X(x_1, x_2; t_1 - t_2) dx_1 dx_2 \\ &= \phi(t_1 - t_2) \end{aligned}$$

Final Result

$$R_{XX}(t_1, t_2) = R_{XX}(t_1 - t_2)$$

3.2 Wide Sense Stationary (WSS)

SSS requires all orders to be time-invariant. WSS is a weaker condition.

Condition 1

$$E[X(t)] = \mu_X = \text{constant}, \quad \forall t$$

Condition 2

$$E[X(t_1)X^*(t_2)] = R_{XX}(\tau), \quad \tau = t_1 - t_2$$

3.3 Relationship

$$\text{SSS} \Rightarrow \text{WSS}, \quad \text{but} \quad \text{WSS} \not\Rightarrow \text{SSS}$$

Gaussian Process Exception

$$\text{WSS} \Rightarrow \text{SSS}$$

4. Ergodic Process

Time Average of a Random Process

(Only heading provided in context; no further derivation present in slides.)

END OF SCRIBE